

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 9

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\ &\quad - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

$$\begin{aligned}q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \\ p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta, t}^2(\mathbf{x}_t))\end{aligned}$$

It is assumed that $\boldsymbol{\sigma}_{\theta, t}^2(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I}$.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta, t}(\mathbf{x}_t)\|^2 \right]$$

Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization

$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t) \end{aligned}$$

Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta,t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2 \right]$$

At every step of the reverse process, we attempt to predict the noise ϵ that was used in the forward diffusion process!

Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta,t}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon) \right\|^2$$

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

Recap of Previous Lecture

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$
$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Note: The objectives of DDPM and NCSN are almost identical; however, their sampling procedures differ:

- NCSN utilizes annealed Langevin dynamics,
- DDPM employs ancestral sampling.

Recap of Previous Lecture

Summary

- Different Markov chains:
 - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
 - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
 - One can generalize to $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- The objectives coincide: ELBO $\equiv$ score-matching.
- The sampling procedures differ:
 - Ancestral sampling in DDPM;
 - Annealed Langevin dynamics for NCSN;
 - Hybrid approaches that combine both updates are possible.

Recap of Previous Lecture

DDPM Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \epsilon$$

Guided Generation

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \end{aligned}$$

Here, $p(\mathbf{y}|\mathbf{x}_t)$ denotes a classifier operating on noisy samples (which must be trained separately).

Recap of Previous Lecture

Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Note: Increasing γ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Recap of Previous Lecture

Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

Recap of Previous Lecture

The previous method requires an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ trained on noisy data. Let's try to avoid this requirement.

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) \\ \nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot (\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)) \\ &= (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y})\end{aligned}$$

Recap of Previous Lecture

Scaled Guided Score Function

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

CFG

1. Introduce the "absence of conditioning" label $\mathbf{y} = \emptyset$.
2. Identify the unguided score function $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$.
3. Train a single model $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \emptyset$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ and $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$.

Outline

- 01 Continuous-Time Normalizing Flows (CNF)
- 02 Continuity Equation for CNF Log-Likelihood
- 03 SDE Basics

Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

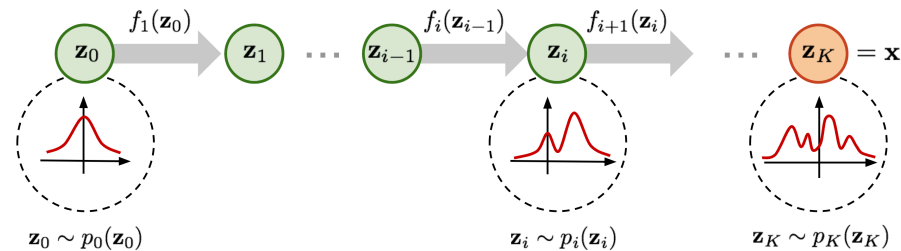
03 SDE Basics

Discrete-Time Normalizing Flows

Change of Variable Theorem (CoV)

Let $\mathbf{x}$ be a random variable with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable and **invertible** transformation. If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{z}) = \mathbf{g}(\mathbf{z})$, then

$$\begin{aligned} p(\mathbf{x}) &= p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| \\ &= p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right| \end{aligned}$$



$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^K \log \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right|.$$

Towards Continuous-Time Normalizing Flows

Up to this point, we have considered discrete-time normalizing flows:

$$\mathbf{x}_{t+1} = \mathbf{f}_\theta(\mathbf{x}_t, t); \quad \log p(\mathbf{x}_{t+1}) = \log p(\mathbf{x}_t) - \log \left| \det \frac{\partial \mathbf{f}_\theta(\mathbf{x}_t)}{\partial \mathbf{x}_t} \right|.$$

Residual Flows

Here $\mathbf{f}_\theta$ is the NF **bijection**, $\mathbf{v}_\theta$ is the **velocity** (vector field):

$$\mathbf{x}_{t+1} = \mathbf{f}_\theta(\mathbf{x}_t, t) = \mathbf{x}_t + \mathbf{v}_\theta(\mathbf{x}_t, t)$$

Given $\mathbf{y} = \mathbf{f}_\theta(\mathbf{x}, t)$, recover $\mathbf{x}$ by **fixed-point iteration**:

$$\mathbf{x} = \mathbf{y} - \mathbf{v}_\theta(\mathbf{x}, t), \quad \mathbf{x}^{(k+1)} = \mathbf{y} - \mathbf{v}_\theta(\mathbf{x}^{(k)}, t).$$

If $\mathbf{v}_\theta$ has Lipschitz constant $L < 1$ in $\mathbf{x}$, the iteration is contractive. **Banach's theorem** guarantees a unique solution for every $\mathbf{y}$, so $\mathbf{f}_\theta$ is invertible.

Towards Continuous-Time Normalizing Flows

A residual step $\mathbf{x}(t + h) = \mathbf{x}(t) + h \mathbf{v}_\theta(\mathbf{x}(t), t)$ is invertible if $hL < 1$:

$$\frac{\mathbf{x}(t + h) - \mathbf{x}(t)}{h} = \mathbf{v}_\theta(\mathbf{x}(t), t)$$

Taking the limit $h \rightarrow 0$, we obtain continuous-time dynamics.

Continuous-Time Dynamics

Consider an Ordinary Differential Equation (ODE):

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0. \\ \mathbf{x}(t_1) &= \int_{t_0}^{t_1} \mathbf{v}_\theta(\mathbf{x}(t), t) dt + \mathbf{x}_0 \end{aligned}$$

The **velocity field** $\mathbf{v}_\theta : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ need not be contractive or invertible. Under suitable regularity conditions, **uniqueness of ODE solutions guarantees an invertible flow**.

ODEs, Flows and Numerical Solutions

Ordinary Differential Equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t), \quad \mathbf{x}(t_0) = \mathbf{x}_0.$$

Flow

The **flow** $\psi : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ maps the initial point to its position at time t : $\mathbf{x}(t) = \psi_t(\mathbf{x}_0)$.

$$\frac{d\psi_t(\mathbf{x}_0)}{dt} = \mathbf{v}_{\theta}(\psi_t(\mathbf{x}_0), t), \quad \psi_{t_0}(\mathbf{x}_0) = \mathbf{x}_0.$$

Numerical Solution

$$\psi_t(\mathbf{x}_0) = \mathbf{x}_0 + \int_{t_0}^t \mathbf{v}_{\theta}(\mathbf{x}(s), s) ds \approx \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t).$$

A numerical solver approximates the flow by a sequence of **finite steps**.

Numerical Solution of ODEs: Euler

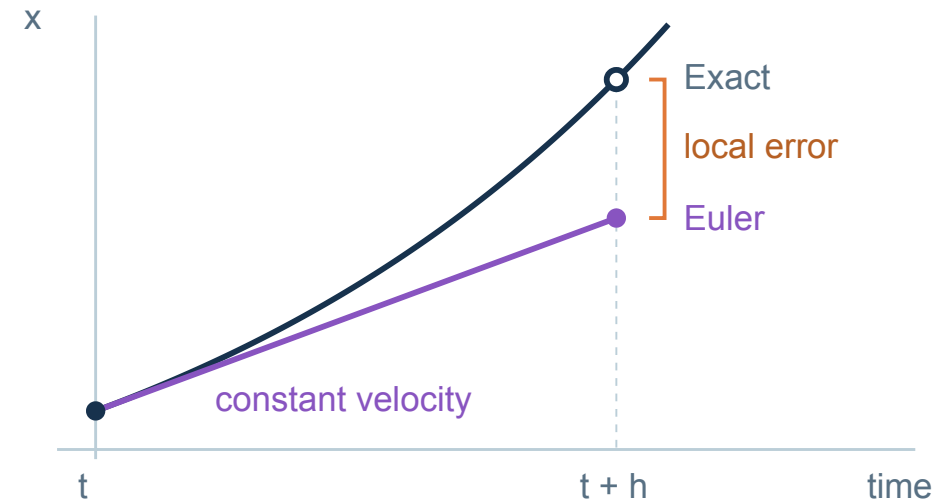
Euler Update Step

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_\theta(\mathbf{x}(t), t)$$

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_\theta(\mathbf{x}(t), t)$$

Freeze the current velocity over the step: **one field evaluation**.

One step: follow the tangent



One-Step Error

For a smooth field, Taylor expansion from the **exact starting point** gives

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \mathbf{v}_\theta(\mathbf{x}(t), t) + \mathcal{O}(h^2).$$

Euler drops the higher-order terms: the **local error is $\mathcal{O}(h^2)$** .

Numerical Solution of ODEs: Euler and Heun

Same initial point and step size: 8 steps on $[0, 3]$, $h = 0.375$.

Euler

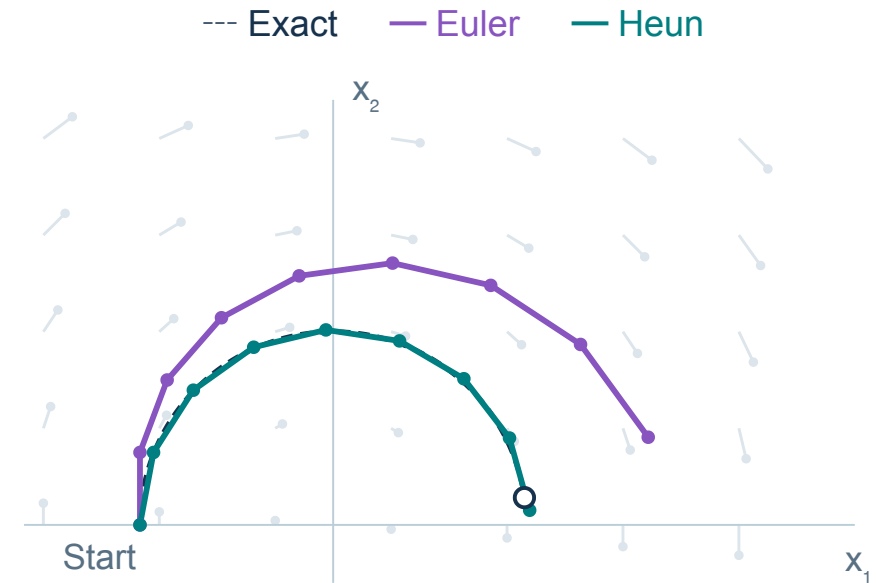
$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

Heun: predict the endpoint

$$\mathbf{x}'(t+h) = \mathbf{x}(t) + h \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

Heun: average the two velocities

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \frac{h}{2} (\mathbf{v}_{\theta}(\mathbf{x}(t), t) + \mathbf{v}_{\theta}(\mathbf{x}'(t+h), t+h))$$



Error 0.713

Error 0.071

Step 8 / 8 · Field evaluations: Euler 8, Heun 16.

Heun averages two velocities.
Twice the field evaluations per step.

Example: $\mathbf{v}(\mathbf{x}) = (x_2, -x_1)$, $\mathbf{x}(0) = (-1, 0)$, $t \in [0, 3]$.

ODE Solvers: Accuracy and Cost

For a sufficiently smooth velocity field on a fixed interval $[0, T]$:

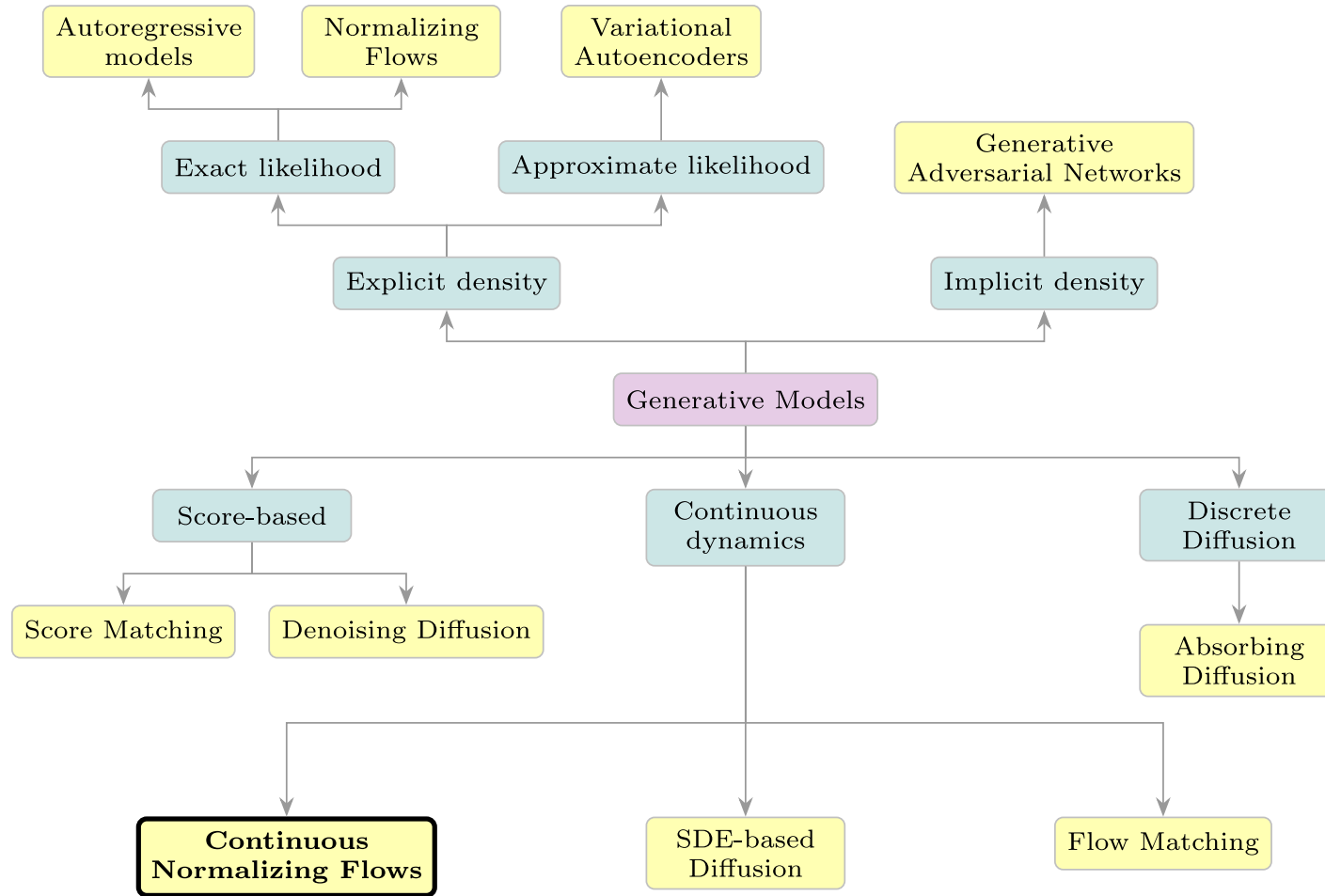
$$\|\widehat{\mathbf{x}}_N - \mathbf{x}(T)\| = \mathcal{O}(h^p), \quad h = T/N.$$

This is the **global error** after N steps.

Method	Global error	Velocity evaluations / step
Euler	$\mathcal{O}(h)$	1
Heun	$\mathcal{O}(h^2)$	2
RK4	$\mathcal{O}(h^4)$	4

- **One-step error** from an exact starting point is $\mathcal{O}(h^{p+1})$.
- For small h , halving the step roughly reduces global error by **2 / 4 / 16**, but doubles the number of steps.
- Compare computational cost by **NFE**: the number of velocity-network evaluations.

Generative Models Taxonomy



Continuous-Time Normalizing Flows: Neural ODE

Neural ODE

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Euler ODESolve

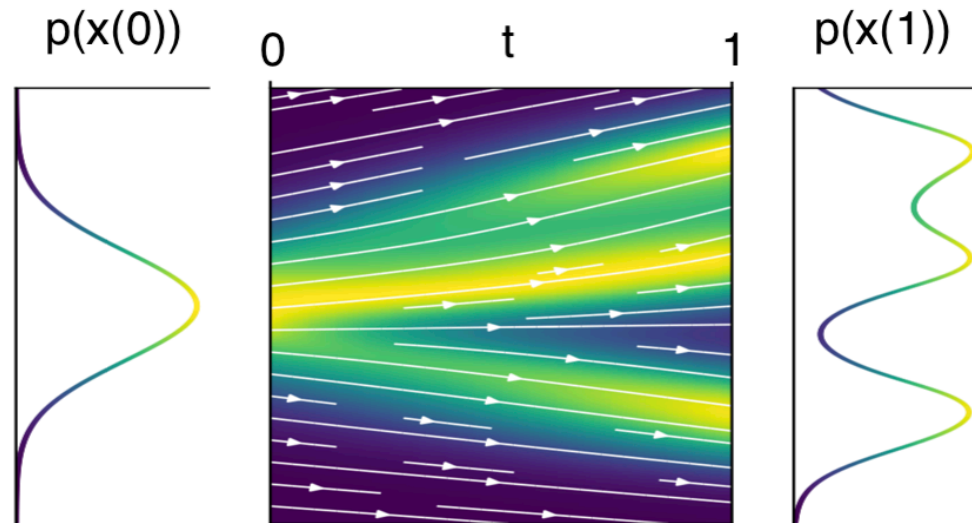
$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

- Consider $[t_0, t_1] = [0, 1]$ for simplicity.
- If $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$,
- Then, for any t , $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.

Continuous-Time Normalizing Flows: Intuition

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** interpolating between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- What is the difference between $p_t(\mathbf{x}(t))$ and $p_t(\mathbf{x})$?



Continuous-Time Normalizing Flows: Reversibility

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t , then the ODE has a **unique solution** given by a flow ψ_t .

This guarantees the ODE is **uniquely reversible**.

$$\psi_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_\theta(\psi_t(\mathbf{x}_0), t) dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

Note: Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection).

How can we compute $p_t(\mathbf{x})$ at arbitrary t ?

Outline

- 01 Continuous-Time Normalizing Flows (CNF)
- 02 Continuity Equation for CNF Log-Likelihood**
- 03 SDE Basics

Continuous-Time NF

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t , and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = - \operatorname{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

Integrating gives the density **along the trajectory**:

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

Hutchinson's trace estimator:

$$\operatorname{tr}(\mathbf{J}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} [\boldsymbol{\epsilon}^T \mathbf{J} \boldsymbol{\epsilon}], \quad \mathbf{J} = \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}.$$

- One noise vector gives an **unbiased trace estimate** via autodiff, without forming the full Jacobian.
- **Exact identity; approximate computation** using an ODE solver and a stochastic trace estimate.

Divergence: Where Does the Density Go?

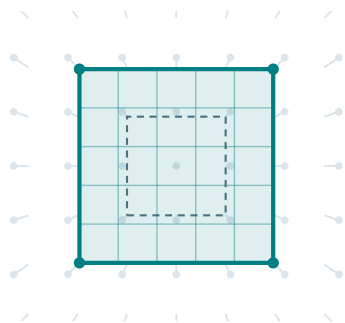
Lecture 2: area change

$$\frac{A_t}{A_0} = |\det \mathbf{J}_{\psi_t}|, \quad \mathbf{J}_{\psi_t} = \frac{\partial \psi_t}{\partial \mathbf{x}_0}$$

CNF: log-area rate

$$\frac{d}{dt} \log |\det \mathbf{J}_{\psi_t}| = \operatorname{div} \mathbf{v} = \operatorname{tr} \left(\frac{\partial \mathbf{v}}{\partial \mathbf{x}} \right)$$

Expansion · $\mathbf{v} = 0.45\mathbf{x}$



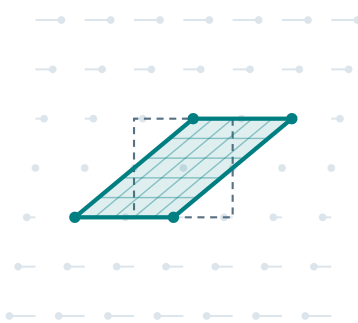
Divergence	0.90
Area	3.857
Density inside the patch	0.259
Area × density	1.000

Contraction · $\mathbf{v} = -0.45\mathbf{x}$



Divergence	-0.90
Area	0.259
Density inside the patch	3.857
Area × density	1.000

Shear · $\mathbf{v} = (0.8x_2, 0)$



Divergence	0.00
Area	1.000
Density inside the patch	1.000
Area × density	1.000

Mass is conserved: $\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\frac{d \log A_t}{dt} = -\operatorname{div} \mathbf{v}.$

t = 1.5. Dashed: initial unit square; solid: moving patch. Area scaling is local in general; exact here (linear flows).

CNF: Training and Sampling

Fix $p_0 = \mathcal{N}(0, \mathbf{I})$ and learn $\mathbf{v}_\theta$; the endpoint density is $p_1 = p_\theta$.

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}$. Solve the ODE **backward**, from $\mathbf{x}(1) = \mathbf{x}$ to $\mathbf{x}(0)$.

2. Compute the negative log-likelihood along this trajectory:

$$\mathcal{L} = -\log p_\theta(\mathbf{x}) = -\log p_0(\mathbf{x}(0)) + \int_0^1 \text{tr} \left(\frac{\partial \mathbf{v}_\theta(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

3. Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$; differentiate through the ODE, e.g., via the **adjoint method**.

Sampling

1. Sample $\mathbf{x}(0) \sim p_0$.

2. Solve the ODE **forward** and return the endpoint:

$$\mathbf{x}(1) = \text{ODESolve}_v(\mathbf{x}(0), \theta, t_0 = 0, t_1 = 1) \sim p_\theta.$$

Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

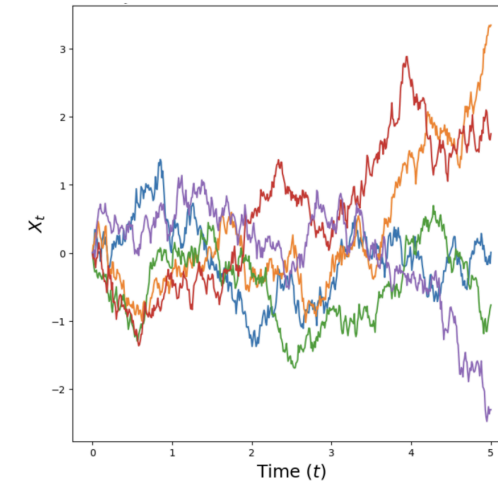
03 SDE Basics

Stochastic Differential Equation (SDE)

Wiener Process

$\mathbf{w}(t)$ is the standard Wiener process (Brownian motion), defined by:

1. $\mathbf{w}(0) = 0$ (almost surely);
2. $\mathbf{w}(t)$ has independent increments;
3. $\mathbf{w}(t)$ trajectories are continuous;
4. $\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t - s)\mathbf{I})$ for $t > s$;



$$d\mathbf{w} = \mathbf{w}(t + dt) - \mathbf{w}(t) = \mathcal{N}(0, \mathbf{I} \cdot dt) = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Stochastic Differential Equation (SDE)

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t) \quad \Rightarrow \quad d\mathbf{x} = \mathbf{v}(\mathbf{x}, t) dt$$

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

- $\mathbf{f}(\mathbf{x}, t) : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^m$ is the **drift** term (vector field).
- $g(t) : \mathbb{R} \rightarrow \mathbb{R}$ is the **diffusion** term (if $g(t) = 0$, we recover the standard ODE).
- $\mathbf{w}(t)$ is the standard Wiener process ($d\mathbf{w} = \epsilon \cdot \sqrt{dt}$).
- We do not have the flow $\psi_t(\mathbf{x}_0)$ notion anymore, since trajectories are stochastic.

Itô Integration Basics

The SDE should be understood in its **integral form** (Itô sense):

$$\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mathbf{f}(\mathbf{x}(s), s) ds + \int_0^t g(s) d\mathbf{w}(s)$$

- The **first integral** is a classical (Riemann) integral.
- The **second integral** is an **Itô stochastic integral**.

$$\int_0^t g(s) d\mathbf{w}(s) = \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} g(t_i) (\mathbf{w}(t_{i+1}) - \mathbf{w}(t_i))$$

- Brownian paths are continuous but a.s. **nowhere differentiable** $\Rightarrow$ Fundamental Theorem of Calculus fails.
- **Itô's lemma** is the stochastic counterpart of the chain rule; we will not need its explicit form here.
- Expressions $d\mathbf{x}$, dt , $d\mathbf{w}$ are **formal shorthand** for infinitesimal increments.

Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

Theorem

If $\mathbf{f}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t and $g(t)$ is continuous then the SDE has the solution given by unique process $\mathbf{x}(t)$.

- Unlike ODEs, the initial condition $\mathbf{x}(0)$ doesn't uniquely determine the trajectory.
- There are two sources of randomness:
 - the initial distribution $p_0(\mathbf{x})$;
 - the Wiener process $\mathbf{w}(t)$.

Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

Discretizing the SDE (Euler-Maruyama Update) – SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

If $dt = 1$, then

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{f}(\mathbf{x}_t, t) + g(t) \cdot \epsilon$$

- At any time t , the process has density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$ specifies a **probability path** from $p_0(\mathbf{x})$ to $p_1(\mathbf{x})$.
- How can we obtain the probability path $p_t(\mathbf{x})$ for $\mathbf{x}(t)$?

Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t) dt + g(t) d\mathbf{w}, \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Here,

$$\operatorname{div}(\mathbf{v}) = \sum_{i=1}^m \frac{\partial v_i(\mathbf{x})}{\partial x_i} = \operatorname{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x})}{\partial \mathbf{x}} \right)$$

$$\Delta_{\mathbf{x}}p_t(\mathbf{x}) = \sum_{i=1}^m \frac{\partial^2 p_t(\mathbf{x})}{\partial x_i^2} = \operatorname{tr} \left(\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

Stochastic Differential Equation (SDE)

Theorem (Kolmogorov-Fokker-Planck)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

- The KFP theorem is a necessary and sufficient condition (it uniquely defines $p_t(\mathbf{x})$).
- This generalizes the continuity equation for continuous-time NF:

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right).$$

Special Case: constant density $p_t(\mathbf{x})$

Let's find the SDE for which $p_t(\mathbf{x}) = \text{const}$ (i.e., if $\mathbf{x}(0) \sim p_0(\mathbf{x})$, then $\mathbf{x}(t) \sim p_0(\mathbf{x})$).

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = 0$$

Langevin SDE (Special Case)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = 0$$

$$\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] = \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}$$

$$\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x}) = \frac{1}{2} g^2(t) \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}$$

$$\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} g^2(t) \frac{1}{p_t(\mathbf{x})} \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} = \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

Let $g(t) = 1$, then $\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

Langevin SDE (Special Case)

Let's find the SDE for which $p_t(\mathbf{x}) = \text{const}$ (i.e., if $\mathbf{x}(0) \sim p_0(\mathbf{x})$, then $\mathbf{x}(t) \sim p_0(\mathbf{x})$).

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

Discretized Langevin SDE

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \frac{\eta}{2} \cdot \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_\theta(\mathbf{x})$ for all t :

Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

We (partially) explained, why Langevin dynamics is working.

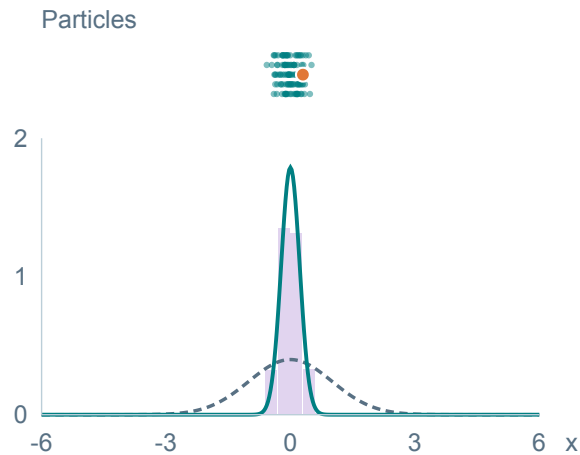
Langevin: Moving Particles, Stationary Density

$$p_0(x) = p_*(x) = \mathcal{N}(0, 1), \quad \frac{1}{2} \nabla_x \log p_*(x) = -\frac{1}{2}x$$

Same initial samples from the target. All three cases at $t = 3$.

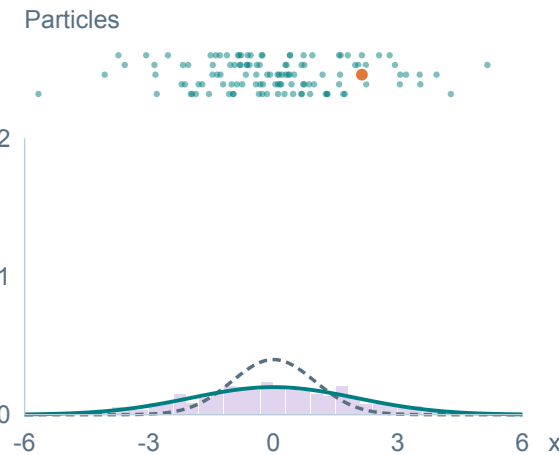
▤ Empirical histogram — Exact current density --- Target $\mathcal{N}(0, 1)$

Drift only · $dx = -\frac{1}{2}x dt$



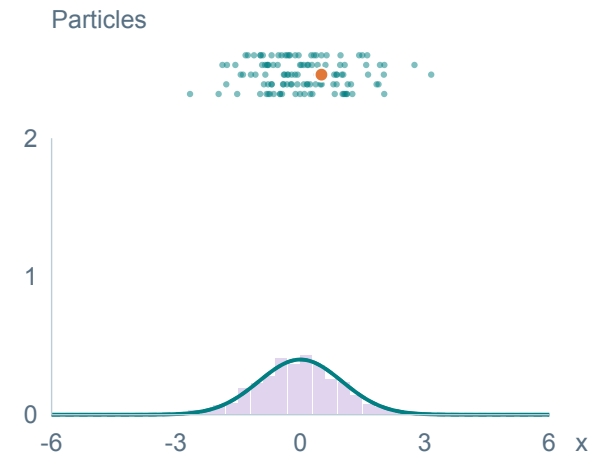
Exact variance: 0.050

Diffusion only · $dx = dw$



Exact variance: 4.000

Both · $dx = -\frac{1}{2}x dt + dw$



Exact variance: 1.000

Stationary density does not mean stationary particles.

768 samples; finite-sample fluctuations remain. Exact Gaussian transitions isolate stationarity from discretization error.

Summary

- CNFs define invertible transformations via neural ODEs; likelihood-based training requires ODE solves, trace evaluations and gradients through the solver.
- If $\mathbf{x}_0$ is a random variable, this yields a **probability path** $p_t(\mathbf{x})$ as time evolves. The continuity equation describes the evolution of $\log p(\mathbf{x}, t)$ over time.
- An SDE defines a stochastic process with drift and diffusion terms; ODEs are a special case of SDEs.
- The KFP equation describes the probability dynamics of an SDE.
- The Langevin SDE preserves a constant probability path.